The Golden Split: Icosahedra, Unborns and AXE Become Two Convex Space-Fillers



Theorem: in the simple cubic lattice of icosahedra, cutting every AXE from the point 1/φ of the way along its axis, and giving each of its four pyramids to the neighbour it touches, turns the icosahedra and the Unborns into two convex truncated octahedra, the Icosaxes and the Unborn Axes, which fill space; this is the only cut point at which both are convex, and in the Lotus Jewel network the same cut gives a keyed pair, the Lotus Jewel Axes and the Dodecaxes, of volumes √5 and 2

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Paper #48 · Version 2 · 11 November 2026 (Version 1: 23 October 2026) · DICTO DOI: 10.5281/zenodo.23596362 · Licence CC BY-SA 4.0 · Scripts, exact checks and figures (Shadow forms): https://github.com/DICTOR-Master/kaleidohedra/tree/066a46a/studies/2026-10-23-shadow-forms (research branch, commit 066a46a; also attached to this record)
The ideas are DICTO's: the Lotus Jewel network and the AXE that fills its gap, the "shadow forms" (other ways of breaking up the same space to find new pieces), the aim of unit pieces without odd protrusions, the polar splits "hemispherical, or anywhere in between", and the names Unborn, AXE, Lotus Jewel, Icosaxes, Unborn Axes, Lotus Jewel Axes and Dodecaxes; computation, exact proofs and drafting with Claude (Anthropic).

Version 2 adds the permutohedron reading (no result changes): T(x), the hull of the signed orders of (1 + x, x, 0), is the permutohedron of four values placed symmetrically with gaps x : 1 : x (#105, Theorem 4): Kelvin's cell at equal gaps, the golden pair at x = τ, and the DICTO 1:2 and DICTO 2:1 Truncated Octahedra at x = 1/2 and x = 2. Its hexagons are truncated triangles, the permutohedra of three consecutive values, with sides in the ratio x (to appear in DICTO, The Permutohedron Paradigm, Chapter 1).

Abstract. Icosahedra cannot fill space on their own. Set them in a cubic grid, edge to edge, and two kinds of gap are left: a squeezed icosahedron (the Unborn) in the middle of each cube, and thin tetrahedra (the AXE) in the faces between them, six per cube. The AXE is the awkward one: wherever it is glued, it sticks out. We cut each AXE into four little pyramids from one inner point and hand each pyramid to the piece it touches. At exactly one point, 1/φ of the way along the AXE's axis, the golden point, nothing sticks out any more: the icosahedra and the Unborns both become truncated octahedra, relatives of Kelvin's foam cell, and these two fill space with no gap.
Main theorem. In the simple cubic lattice of regular icosahedra of edge 1 and spacing φ, cone every AXE (the gap tetrahedron, tile t6 of [3] scaled by 1/τ) from the point 1/φ of the way from the midpoint of its long edge to the midpoint of its short edge, and give each of the four pyramids to the piece whose face it lies on. Then the icosahedron with its 12 pyramids, the *Icosaxes*, and the Unborn with its 12, the *Unborn Axes*, are convex truncated octahedra with full octahedral symmetry, the same equiangular hexagon (sides alternately 1/√2 and 1/(φ√2)) and squares of side 1/√2 and 1/(φ√2); their volumes are 1 + 2√5/3 and 1 + √5/3, and they fill space face to face, Icosaxes at the lattice points and Unborn Axes at the cube centres. **Theorem 2:** for any cut points whatever, inside the AXEs, both padded pieces are convex only if every AXE is cut at its golden point.
Theorem 3: in the Lotus Jewel network the same cut gives the *Lotus Jewel Axes* (8 dents, volume √5) and the *Dodecaxes* (8 keys, volume 2), and every way of giving each key to one side or the other still fills space. **Theorem 4 (factor x):** the pair belongs to a one-

parameter family of mixed space-fillers, the semi-uniform truncated octahedra T(x) and T(1/x) (Koca's orbit [1, eq. 49]) at the corners and centres of a cubic lattice, from Kelvin's cell (x = 1) to the golden pair (x = τ). All results are exact. The two smooth shapes are known: they are Koca et al.'s non-regular truncated octahedra x = τ and x = 1/τ [1] (the semi-uniform truncated octahedron [4]); what is new, in the sources we checked, is the route from the icosahedron lattice, the golden split and its uniqueness, and the keyed pair.

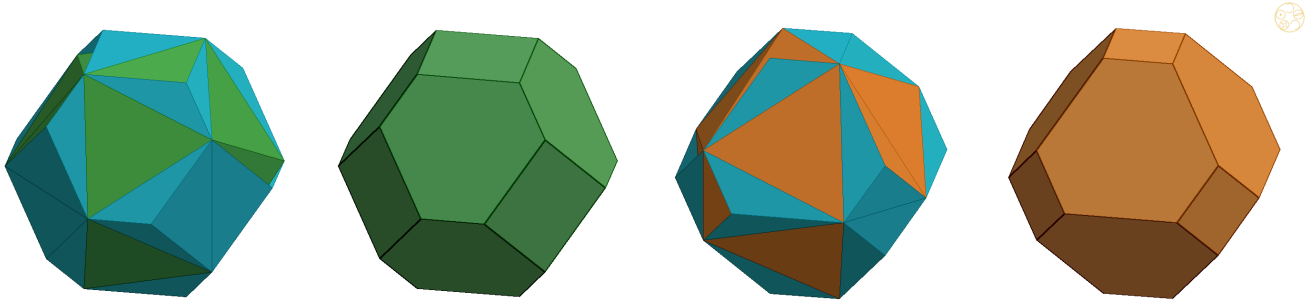


Figure 1. The golden pair. From left: the icosahedron (green) with its 12 AXE pyramids (cyan); the Icosaxes as one solid; the Unborn (orange) with its 12 AXE pyramids; the Unborn Axes as one solid. Each panel is drawn at its own scale. Both are truncated octahedra with full octahedral symmetry.

1. Where it started

> *DICTO had a network of icosahedra with a gap he could fill exactly, but the filling pieces stuck out wherever they were glued. He proposed pieces with no odd protrusions, made by cuts "hemispherical, or anywhere in between".*

The search behind this paper began with mixtures of the Platonic solids. In *Every Mix of the Platonic Solids Fills Space* (paper #47 [9]) DICTO proposed that the five Platonic solids, in any combination, can be joined into a filling of space, and we proved it for all 31 mixtures, at one edge length, with seven connecting pieces. That reached the aim; but DICTO conjectured that a greater elegance and simplicity was within reach, fewer and cleaner pieces in which the solids themselves do the work, although at that time it was beyond our grasp. This paper gives an answer to that conjecture for the two Platonic solids with five-fold symmetry: for the icosahedron, whole and convex, and, in the keyed form, for the dodecahedron (here of edge 1/φ, while #47 kept one edge length). In DICTO's view, these solids are the most elegant solution he had imagined: one golden cut, and two convex pieces, one holding a whole regular icosahedron at its core; and in the keyed pair, a piece holding a whole regular dodecahedron.

On 10 October 2026 DICTO built the icosahedral octet and its Lotus Jewel, the "clawed" icosahedron (Kaleidohedra DISCOVERIES F21). On 23 October he conjectured that the Lotus Jewel network fills space, and proposed finding the piece that fills its gap; the answer was the AXE, exactly six per cubic cell, and with it three exact fillings of the simple cubic icosahedron lattice (the AXE gap study [8]): icosahedra + Unborns + AXE; Lotus Jewels + dodecahedra + AXE; and Lotus Jewel + Tricap + Unborn + AXE kept apart, with any choice of giving each Tricap to one side or the other. The same day DICTO proposed looking for "shadow forms", complementary ways of breaking up the same space, to create new pieces. Fusing neighbouring pieces gave chunky shapes, but every fusion with an AXE kept a spike; giving each AXE whole to one icosahedron fills space with only two kinds of piece, but leaves the icosahedron 68% of its hull, a star. DICTO then set the aim, "unit pieces without odd

protrusions", and the tool, polar splits "hemispherical really, or anywhere in between". Cutting the AXE itself, from one inner point, gave the result of this paper.

2. The network

> *Regular icosahedra sit at the corners of a cubic grid, touching their six neighbours along an edge. Between them are two kinds of hole, and both are filled exactly by three pieces.*

Let $\varphi = (1 + \sqrt{5})/2 = \tau$. Place a regular icosahedron I of edge 1 at every point of the lattice $\varphi\mathbb{Z}^3$, with corners $(0, \pm\tfrac{1}{2}, \pm\varphi/2)$ and their cyclic permutations; each icosahedron shares one edge with each of its six neighbours along the axes. The cell $\varphi^3 = 2 + \sqrt{5}$ holds, besides its icosahedron (volume $(15 + 5\sqrt{5})/12$):

- at the cube centre $c = (\varphi/2, \varphi/2, \varphi/2)$, the **Unborn** U, with corners $c + (\pm\varphi/2, \pm 1/(2\varphi), 0)$ and cyclic permutations: 12 corners, 24 edges of length 1 and 6 of length $1/\varphi$, 8 equilateral faces and 12 isosceles faces $(1, 1, 1/\varphi)$, volume $(3 + 7\sqrt{5})/12$. It is the pseudoicosahedron of Koca et al. [1, eq. 39] with $x = 1/\tau$, whose corners, as they note, are the 12 corners of a dodecahedron that are not on its cube; the name is DICTO's.
- six **AXE**, the Mosseri–Sadoc tile t6 in the numbering of [3] (see also [2]), scaled by $1/\tau$: a tetrahedron with five edges 1 and one edge $1/\varphi$, volume 1/12. Its *long edge* (the edge of length 1 opposite the short one) is the edge two neighbouring icosahedra share, its *short edge* ($1/\varphi$) an edge two neighbouring Unborns share; of its four faces, the two through the long edge lie on the two icosahedra, the two through the short edge on the two Unborns.

The icosahedron meets the Unborn along the Unborn's 8 equilateral faces. The Unborn is the regular dodecahedron D of edge $1/\varphi$ centred at c with its 8 cube corners cut off: each corner of D that reaches into an icosahedron is a **Tricap** (tile t3 of [3], scaled by $1/\tau$; volume $(3 - \sqrt{5})/24$), the tetrahedron $D \cap I$ standing on one of the icosahedron's faces that meet the Unborn. Giving the 8 Tricaps to the dodecahedron instead of the icosahedra changes the network: the icosahedron minus its 8 Tricaps is the **Lotus Jewel** J, volume $(1 + 3\sqrt{5})/4$, and the Unborn plus its 8 Tricaps is D, volume $(15 + 3\sqrt{5})/12$. Per cell, $I + U + 6\,\mathrm{AXE} = J + D + 6\,\mathrm{AXE} = \varphi^3$ exactly, and both fill space with no gap; so does every assignment of the Tricaps [8].

3. The golden split

> *Pick one point inside each AXE and cut the AXE into four pyramids, each with one AXE face as its base. Give each pyramid to the piece its base touches. The AXE is gone; the question is only where to put the point.*

For an AXE with long edge AB and short edge CD let m_1, m_2 be their midpoints, and for $0 \le t \le 1$ let $p(t) = m_1 + t(m_2 - m_1)$, a point of the AXE's axis (the line through m_1 and m_2 is its two-fold axis). The *split at t* replaces the AXE by the four pyramids with apex p(t) over its faces; the two over ABC and ABD join the icosahedra those faces lie on, the two over CDA and CDB the Unborns. Every icosahedron touches 12 AXE faces and every Unborn 12, so each gains 12 pyramids; the pieces still fill space, since only the owner of each part of each AXE has changed. Write I(t) and U(t) for the padded pieces. The **golden split** is $t = 1/\varphi$.

> **Main theorem (the golden split).** At $t = 1/\varphi$:
> (a) the **Icosaxes** $I(1/\varphi)$ is convex: it is the truncated octahedron with corners all permutations of $(\pm\tfrac{1}{2}, \pm\varphi/2, 0)$, with 8 hexagons and 6 squares;

(b) the **Unborn Axes** U(1/φ) is convex: it is the truncated octahedron with corners c + all permutations of (±1/(2φ), ±φ/2, 0), with 8 hexagons and 6 squares;
(c) both have full octahedral symmetry; all 16 hexagons are congruent, equiangular (120°), with sides alternately 1/√2 and 1/(φ√2); the squares of the Icosaxes have side 1/√2, those of the Unborn Axes 1/(φ√2);
(d) their volumes are 1 + 2√5/3 and 1 + √5/3, together φ^3;
(e) the 12 apices of the Icosaxes are the corners of the mirror image of its icosahedron in the plane x = y through its centre, and the 12 apices of the Unborn Axes those of the mirror image of its Unborn in the plane x = y through c;
(f) Icosaxes at the points of $\varphi\mathbb{Z}^3$ and Unborn Axes at the cube centres fill space face to face: Icosaxes meet Icosaxes on their squares, Unborn Axes meet Unborn Axes on theirs, and the two meet each other on the hexagons.

> *Why it is true: every corner of every piece is a sum of halves and golden numbers, so everything can be checked with exact arithmetic. Convexity is checked by comparing two volumes: the pieces glued together, and the smallest convex solid around them.*

Proof. All coordinates lie in $\mathbb{Q}(\sqrt{5})$, and the study's check [8a] works in that field with no floating point. (a), (b): the padded piece is the union of the core and its 12 pyramids, which have disjoint interiors (the AXEs and the cores do, in the filling [8]), so its volume is the sum of their volumes, exact determinants; its convex hull is found exactly (every plane through three of its points with all points on one side), and its volume computed exactly. The two volumes are equal, (15 + 5√5)/12 + 12 pyramids = 1 + 2√5/3 for the Icosaxes and (3 + 7√5)/12 + 12 pyramids = 1 + √5/3 for the Unborn Axes, so the union, a closed set inside its hull with the same volume, is the hull, which is convex. The hull has 24 corners and 14 faces, 8 with six corners and 6 with four, and its corner set is exactly the stated orbit, which is invariant under all permutations and sign changes of the coordinates: hence (c)'s symmetry. The side lengths and the congruence of the hexagons (all 15 distances between their corners) are compared exactly. (d) is the sum above. (e): the 12 apices p(1/φ) are compared, as a set, with the core's corners under (x, y, z) ↦ (y, x, z). (f): the pieces fill space because the split only re-assigns parts of the AXE in a filling [8]; the shared faces are found by comparing corner sets: the hexagon of an Icosaxes towards (1, 1, 1) is, corner for corner, the hexagon of the Unborn Axes at c facing back, and the square of an Icosaxes at x = φ/2 is the square of its neighbour at (φ, 0, 0); the others follow by symmetry. The check passes all its 21 tests; a 500,000-point test of the filling (in floating point, as a second check) finds no point uncovered and none covered twice. ∎

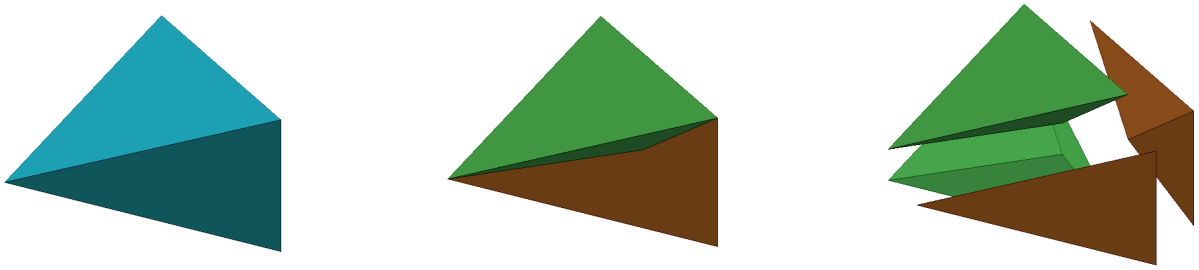


Figure 2. The split of one AXE. Left: the AXE. Middle: cut from the point p(1/φ) of its axis into four pyramids. Right: pulled apart: the two pyramids on its icosahedron faces (green) go to the icosahedra, the two on its Unborn faces (brown) to the Unborns.

4. Only the golden point

> *Move the cutting point a little towards the Unborns and the padded icosahedron grows a ridge; move it the other way and the padded Unborn does. Only at the golden point are both smooth, wherever else in the AXE the point might be tried.*

> **Theorem 2 (uniqueness).** Cut each AXE from any point p inside it, possibly different for different AXEs, and give its four pyramids to their neighbours as before. If the padded icosahedra and the padded Unborns are all convex, then every AXE is cut at its golden point p(1/φ). In particular, for cut points p(t) on the axis ($0 \le t \le 1$) the padded icosahedron I(t) is not convex for $t > 1/\varphi$ and the padded Unborn U(t) is not convex for $t < 1/\varphi$.

> *Why: where an icosahedron meets an Unborn, their shared face gets no pyramid, so it stays on the outside of both pieces. A convex piece cannot have anything poking out past one of its own faces, so the cutting point must stay on the icosahedron's side of that face and also on the Unborn's side: it must lie exactly in it. Each AXE touches four such faces, and they meet in just one point, the golden point.*

Proof. An AXE's four pyramids go to two icosahedra I_0, I_1 and two Unborns U_0, U_1, and all four have the same apex p. Each pair I_i, U_j shares a face (one of the icosahedron's 8 faces that meet Unborns), which carries no pyramid and so stays on the boundary of both padded pieces. If the padded I_i is convex, p lies on its side of that face's plane; if the padded U_j is convex, on the other side; so p lies on the plane. For the AXE with corners $(\mp½, \varphi/2, 0)$, $(0, \varphi/2 \pm 1/(2\varphi), \varphi/2)$ the four planes are $x + y + z = \varphi^2/2$, $-x + y + z = \varphi^2/2$, $x - y + z = -1/(2\varphi)$ and $-x - y + z = -1/(2\varphi)$; they have rank 3 and meet in the single point $(0, \varphi/2, ½) = p(1/\varphi)$ (checked exactly in [8a]; by symmetry the same holds for every AXE). For the second statement: the icosahedron's 8 faces that meet Unborns stay on the boundary of I(t), so if I(t) were convex, every apex would lie on the inner side of their planes. The apex p(t) moves linearly in t, so its signed distance from each such plane is $d_0 + t \cdot d_1$ with exact d_0, d_1. For 24 pairs (apex, face) this distance is zero exactly at $t = 1/\varphi$, with $d_1 > 0$: for $t > 1/\varphi$ those apices are outside, and I(t) is not convex. The same holds for the Unborn's 8 equilateral faces, which carry no pyramid, with $d_1 < 0$: U(t) is not convex for $t < 1/\varphi$. With the Main theorem, both are convex at $t = 1/\varphi$. ∎ (In floating point, at every sampled t, I(t) is convex for $t \le 1/\varphi$ and U(t) for $t \ge 1/\varphi$, Table 1.)

t	0	0.2	0.4	0.6	1/φ	0.7	0.8	1
padded icosahedron I(t): fill of hull	100%	100%	100%	100%	**100%**	93.6%	86.9%	76.0%
padded Unborn U(t): fill of hull	58.2%	67.6%	79.8%	97.9%	**100%**	100%	100%	100%

Table 1. How much of its convex hull each padded piece fills, as the cutting point moves from the long edge (t = 0: the Unborns take the whole AXE) to the short edge (t = 1). 100% means convex. Floating point (study script axesplit1.py); the exact statement is Theorem 2.

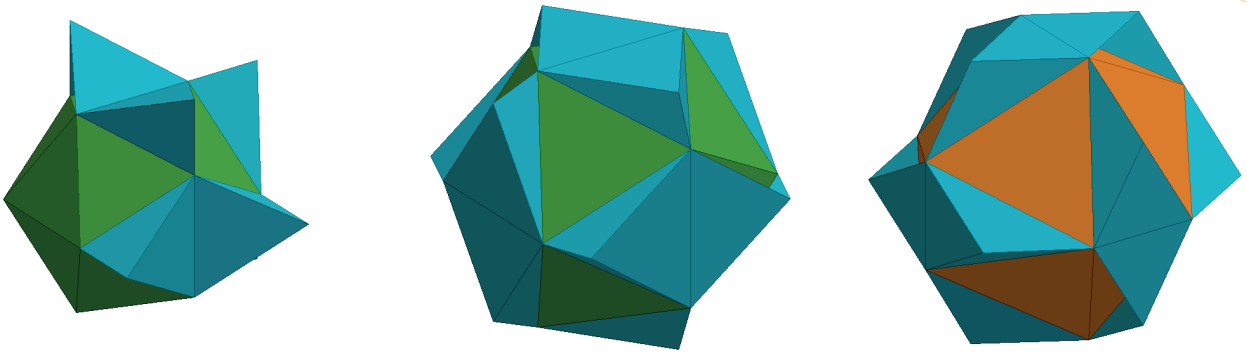


Figure 3. What sticks out when the cut is not golden. Left: an icosahedron given six whole AXE (cyan) — the pieces fill space, but each is a star filling only 68.2% of its hull. Middle: the padded icosahedron I(0.85), cut past the golden point: the pyramids (cyan) rise above the icosahedron's free faces (green) in ridges; 83.9% of its hull. Right: the padded Unborn U(0.35), cut short of it: the Unborn's free faces (orange) sink between the pyramids; 76.4% of its hull. At t = 1/φ both are 100% (Figure 1).

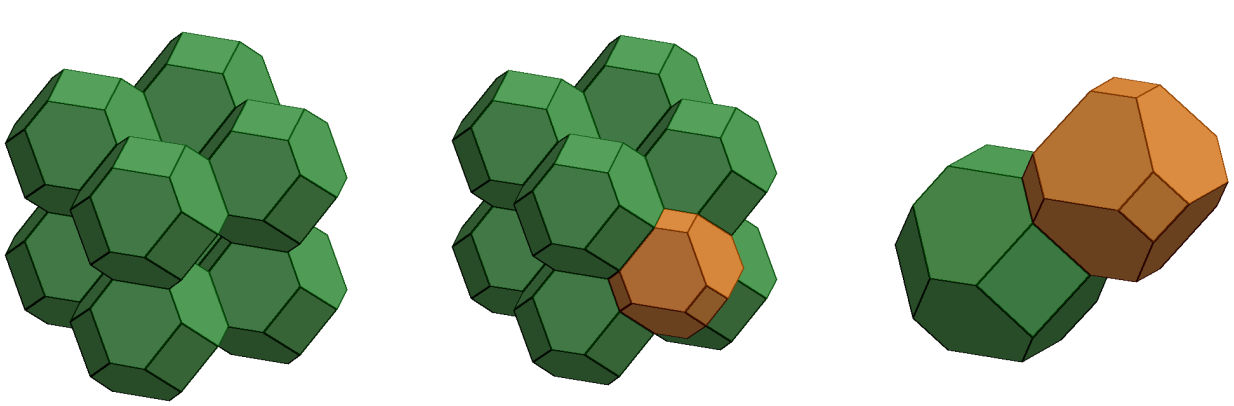


Figure 4. The filling. Left: eight Icosaxes (green) around one cubic cell; the Unborn Axes in its middle is hidden. Middle: with two Unborn Axes (orange). Right: an Icosaxes and an Unborn Axes meeting on a hexagon.

5. The keyed pair

> *In the Lotus Jewel network the eight Tricaps belong to the dodecahedra, not the icosahedra. After the golden split they show up as eight keys on the small piece and eight matching dents in the big one.*

> **Theorem 3 (Lotus Jewel Axes and Dodecaxes).** In the Lotus Jewel network, the golden split gives the **Lotus Jewel Axes** J + 12 pyramids, which is the Icosaxes with its 8 Tricaps removed, and the **Dodecaxes** D + 12 pyramids, which is the Unborn Axes with 8 Tricaps added. Their volumes are exactly √5 and 2. Each Tricap stands on a hexagon shared by an Icosaxes and an Unborn Axes; for every choice of giving each Tricap to the one or the other, the resulting pieces fill space.

> *Why: the Tricaps sit on the faces where icosahedron and Unborn meet, and the AXE pyramids sit on other faces, so the two never get in each other's way. Moving a Tricap from one side to the other only moves volume across a shared face, so the pieces still fill space whichever way each Tricap goes.*

Proof. The AXE faces lie on faces of I and U that share no area with the Tricaps (each Tricap stands on one of the icosahedron's 8 faces that meet Unborns, which carry no pyramid, and touches an AXE face at most along an edge), so the same 12 pyramids join J and D; on D each pyramid stands on part of a pentagon, which is an isosceles face of the Unborn together with two side faces of Tricaps, all in one plane. The volumes are $1 + 2\sqrt{5}/3 - 8(3 - \sqrt{5})/24 = \sqrt{5}$ and $1 + \sqrt{5}/3 + 8(3 - \sqrt{5})/24 = 2$, checked exactly in [8a]. Each Tricap stands on one of the icosahedron's faces that meet Unborns, which lies in a hexagon of the Main theorem. Every assignment of the Tricaps fills space before the split [8]; the split re-assigns only AXE parts, so it fills space after. ∎

So the whole family of the AXE gap reduces to **one golden pair of truncated octahedra in three forms**: smooth (Icosaxes and Unborn Axes), keyed (Lotus Jewel Axes and Dodecaxes), or any mixture, hexagon by hexagon, in which each hexagon is flat or keyed one way or the other. Every such piece is convex apart from its keys and dents, which fit into each other like a three-dimensional jigsaw. The patterns of the Mixed blocks world (layers, rods, checker, pyrochlore) [8] carry over unchanged. Each piece has flat, parallel squares top and bottom.

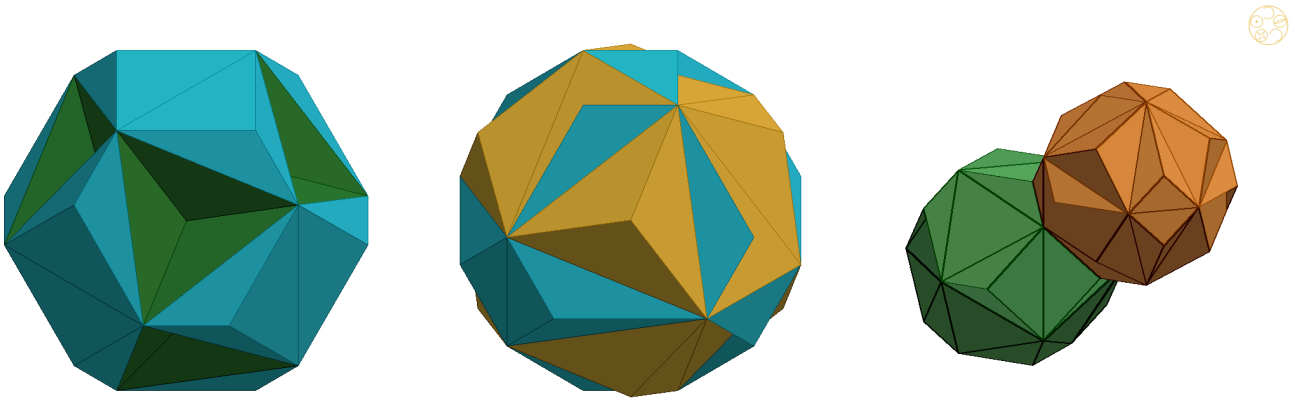


Figure 5. The keyed pair, seen along a three-fold axis. Left: the Lotus Jewel Axes (Lotus Jewel green, AXE pyramids cyan), with a Tricap-shaped dent. Middle: the Dodecaxes (dodecahedron gold), with a Tricap key. Right: the two fitted together, key in dent.

6. Factor x: a family of mixed space-fillers

> *The golden pair is one member of a whole family. Slide one flat cut through every small cube of the lattice and you get a big and a small truncated octahedron that fill space together, for every position of the cut. DICTO calls this position his factor; at the golden value the icosahedron appears inside.*

> **Theorem 4 (factor x).** For $x > 0$ let $T(x)$ be the semi-uniform truncated octahedron [4], written as Koca's orbit [1, eq. 49]: the convex hull of all permutations of $(\pm(1 + x), \pm x, 0)$, scaled so that its squares lie at distance $L/2$ from its centre. Then for every $x > 0$, copies of $T(x)$ at the points of $L\mathbb{Z}^3$ and copies of $T(1/x)$ at the cube centres $(L/2)(1, 1, 1) + L\mathbb{Z}^3$ fill space face to face. Their volumes add to L^3. At $x = 1$ both are the uniform truncated octahedron (Kelvin's cell); at $x = \tau$ they are the Icosaxes and the Unborn Axes ($L = \varphi$).

> *Why: cut space into small cubes, half the lattice spacing on a side. Each small cube has a lattice point at one corner and a cube centre at the opposite corner, and one flat cut across its long diagonal shares it between them. The two truncated octahedra are exactly what these shares add up to.*

Proof. With scale s = L/(2(1 + x)), T(x) is the intersection of the cube $|p_i| \le L/2$ and the octahedron $|p_1| + |p_2| + |p_3| \le k(x)$, where k(x) = (1 + 2x)s = L(1 + 2x)/(2(1 + x)); since L/2 < k(x) < L, its corners are where three of these planes meet, the stated orbit. Likewise T(1/x) has k(1/x) = L(x + 2)/(2(x + 1)), and k(x) + k(1/x) = 3L/2. Space is the union of the small cubes of side L/2 with corners in $(L/2)\mathbb{Z}^3$; each has a lattice point at one end of a long diagonal and a cube centre at the other, and lies in the cube of side L around each of these two and in no other such cube. In the small cube $[0, L/2]^3$ the lattice-point piece is $\{p_1 + p_2 + p_3 \le k(x)\}$ and the centre piece is $\{(L/2 - p_1) + (L/2 - p_2) + (L/2 - p_3) \le k(1/x)\} = \{p_1 + p_2 + p_3 \ge 3L/2 - k(1/x) = k(x)\}$: they share the small cube along one plane, with no gap and no overlap. The same holds in every small cube by symmetry, so the pieces fill space; the plane carries a common hexagon of the two pieces, and the cube faces at distance L/2 carry squares shared with the next piece of the same kind. ∎ (A 300,000-point test agrees for x = 0.3, 0.5, 1/τ, 0.8, 1, 1.25, τ, 2, 3; study script factorx.py.)

So the golden split picks out, in a one-parameter family of mixed space-fillers that are each simple to build, the one member that is also a filling by icosahedra, Unborns and AXE. As a family of weighted Voronoi cells this is not surprising [6]; every T(x) is a pseudoicosahedron and its mirror image [1]; what singles out x = τ (and, with the roles swapped, 1/τ) is that this pseudoicosahedron is the regular icosahedron (Main theorem (e)).

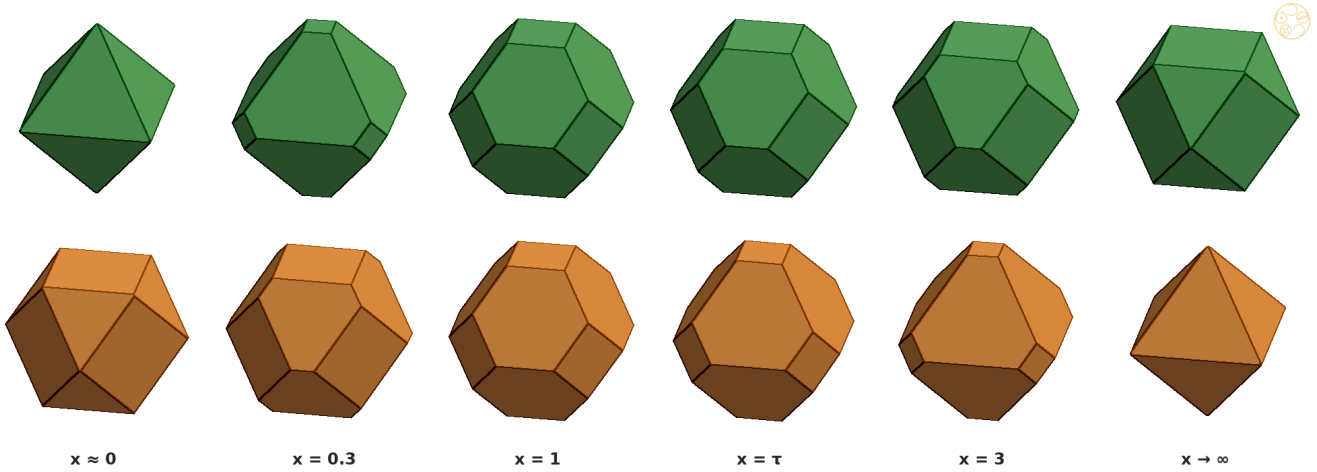


Figure 6. Factor x. Top: T(x), at the lattice points; bottom: T(1/x), at the cube centres; each panel at its own scale. From x ≈ 0 (octahedra with cuboctahedra, the classical rectified cubic honeycomb) through x = 1 (Kelvin's truncated octahedron, both alike) and x = τ (the Icosaxes and the Unborn Axes) to x → ∞ (the two exchanged).

Remark (tetrahedron, octahedron, cube). DICTO proposed the same approach for the tetrahedron, the octahedron and the cube. A first computation (checked exactly, in rational arithmetic, at six values of t; prior work not yet checked) shows the octet, regular tetrahedra and octahedra, inside such a family: cut every double cone from a tetrahedron's centre through a shared face to an octahedron's centre by a plane parallel to the face, a fraction t of the way. At t = 0 the octahedra take everything and are rhombic dodecahedra; for 0 < t < 1/3 small tetrahedra sit between rhombic dodecahedra with their eight three-fold corners cut; t = 1/3 is the octet; for 1/3 < t < 1 small octahedra sit between cubes with four alternate corners

cut; at t = 1 the tetrahedra take everything and are cubes. So the octet lies a third of the way along a road of mixed space-fillers from Kepler's rhombic dodecahedron to the cube. A full treatment is left to a later paper.

7. Conclusion

> *This paper is about mixed space-fillers: two different solids that cannot fill space alone but do fill it together. A lattice of icosahedra leaves two kinds of hole, and the thin AXE pieces between them always stuck out. Cut each AXE at the golden point and share it out, and everything becomes two clean, convex pieces, the Icosaxes and the Unborn Axes, that fill space together. They belong to a whole family set by one number, DICTO's factor, and the golden ratio is the member that holds the icosahedron. The Lotus Jewel version of the same cut gives a pair that locks together with keys.*

It gives an answer, for the icosahedron and, keyed, the dodecahedron, to the conjecture DICTO made after paper #47 [9], that a more elegant and simpler mixed filling was within reach. The golden split answers his two aims at once: unit pieces with no odd protrusions, and a cut "anywhere in between" that is chosen by the geometry itself. Three pieces that cannot fill space alone (icosahedron, Unborn, AXE) become two convex space-fillers with full octahedral symmetry, so icosahedral pieces assemble into a cubic, octahedral whole; the AXE, which the literature uses in aperiodic tilings, here fills a periodic one and then dissolves into it. The keyed forms add a jigsaw on top: every hexagon can be flat or keyed either way, and every such choice fills space; and the factor x turns the pair into a family of mixed space-fillers, from Kelvin’s cell at x = 1 to the golden pair at x = τ. Open questions: the same method for the tetrahedron, octahedron and cube (the octet and its relatives); the golden split in the other AXE networks; the polar splits of the icosahedron and the Unborn themselves; and physical, printed sets of the keyed pieces.

8. What is known, and what is new

> *The two smooth shapes have been described before, as shapes. What we did not find anywhere is how they arise from the icosahedron lattice, the golden cut that makes them, and the keyed pair.*

Known. (i) Koca, Koca, Al-Mukhaini and Al-Qanobi [1, §4.3, eq. 49 in the arXiv numbering] show that the union of a pseudoicosahedron and its mirror image is a non-regular truncated octahedron, the orbit of the octahedral group with Dynkin labels $a_1(1, x, 0)$, that is, all permutations and sign changes of (1 + x, x, 0) up to scale; with their pseudoicosahedron $\pm(1 + x)e_1 \pm x\, e_2$ (cyclic) [1, eq. 39], the Icosaxes is their case x = τ (the regular icosahedron and its mirror image) and the Unborn Axes their case x = 1/τ (the Unborn and its mirror image); Main theorem (e) is this statement for our pieces. (ii) The Polytope Wiki [4] describes the semi-uniform truncated octahedron (8 ditrigons, 6 squares, full octahedral symmetry) and notes that alternating it gives a pyritohedral icosahedron, regular when the ditrigon–ditrigon edge is $(\sqrt{5} - 1)/2$ of the square–ditrigon edge: the Icosaxes. (iii) The regular truncated octahedron is the Voronoi cell of the body-centred cubic lattice (Kelvin's cell; the bitruncated cubic honeycomb [5]); unequal truncated octahedra on a body-centred lattice arise as weighted Voronoi (power, Laguerre) cells [6]; since the Icosaxes and the Unborn Axes share their hexagon with the two edge lengths swapped, their filling in (f) may be seen as a member of that family, and we do not claim the bare tiling. (iv) The AXE and the Tricap are Mosseri–Sadoc tiles (scaled by 1/τ); in the sources we checked they are used in aperiodic, icosahedral

tilings and in dissections of icosahedral solids [2, 3]. (v) As context only: the icosahedron–Unborn network has space group Pm3̄, like the 1/1 approximant α-(AlMnSi), which has icosahedral clusters at the cube corners and centres [7]; the filling by Icosaxes and Unborn Axes has the full Pm3̄m.

Not found, in the sources listed in Appendix A: the simple cubic icosahedron lattice filled by icosahedra, Unborns and AXE turned into these two truncated octahedra; the golden split and its uniqueness (Main theorem and Theorem 2), including the apices landing on the mirror images; and the keyed pair with its free keyings (Theorem 3). "Not found" is limited to those sources; it is not a claim that nobody has seen it.

9. In the apps

The Icosaxes, Unborn Axes, Lotus Jewel Axes and Dodecaxes are being added to Polyhedraverse among DICTO's pieces, with a Kaleidohedra world beside the AXE networks in which the keys can be set by pattern or flipped one hexagon at a time, with X-ray showing the cores and the AXE pyramids (https://dictospheres.com).

References

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[5] "Bitruncated cubic honeycomb", Wikipedia (viewed 23 October 2026).
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[8] DICTO, The AXE gap (study 2026-10-23-axe-gap, Kaleidohedra repository, commit 5f6c20d): the three exact fillings of the simple cubic icosahedron lattice, the Unborn, AXE and Tricap volumes, and every Tricap assignment. [8a] DICTO, Shadow forms (study 2026-10-23-shadow-forms, commit 066a46a), exact/golden_split_exact.py → golden-split-exact.txt.
[9] DICTO, "Every Mix of the Platonic Solids Fills Space", paper #47, DICTO DOI: 10.5281/zenodo.23551326 (2026).

Appendix A. Sources checked for prior work

DICTO's reference library (kaleidohedra-references: manifest, reading list, Fedorov 1885 chapter 13): nothing on these fillings. Read in full: Koca et al. 2015 [1]. Web, 23 October 2026: Polytope Wiki [4]; Koca et al. 2021 [3] and the Mosseri–Sadoc literature (aperiodic only); Wikipedia, truncated octahedron and bitruncated cubic honeycomb [5]; Laguerre and power-diagram tessellation papers [6]; α-(AlMnSi) structure papers [7]; quadibloc, "Filling space with polyhedra"; ProofWiki, "Icosahedron inscribed in octahedron"; searches for two kinds of truncated octahedra on a body-centred lattice, for pyritohedral truncated-octahedron space-fillers, and for icosahedra in a simple cubic lattice with pseudoicosahedral voids. Not checked: Wolfram PolyhedronData, Stella / Great Stella libraries.

Appendix B. Data

Piece	Corners (centred)	Faces	Volume
Icosahedron I	(0, ±½, ±φ/2), cyclic	20 triangles	(15 + 5√5)/12 ≈ 2.181695
Unborn U	(±φ/2, ±1/(2φ), 0), cyclic	8 equilateral, 12 isosceles (1, 1, 1/φ)	(3 + 7√5)/12 ≈ 1.554373
AXE (M–S t6)	e.g. (∓½, φ/2, 0), (0, φ/2 ± 1/(2φ), φ/2)	4 triangles; edges five 1, one 1/φ	1/12
Tricap (M–S t3)	I ∩ D	4 triangles	(3 − √5)/24 ≈ 0.031831
Icosaxes	(±½, ±φ/2, 0), all permutations	8 hexagons, 6 squares (side 1/√2)	1 + 2√5/3 ≈ 2.490712
Unborn Axes	(±1/(2φ), ±φ/2, 0), all permutations	8 hexagons, 6 squares (side 1/(φ√2))	1 + √5/3 ≈ 1.745356
Lotus Jewel Axes	Icosaxes minus 8 Tricaps	keyed (8 dents)	√5 ≈ 2.236068
Dodecaxes	Unborn Axes plus 8 Tricaps	keyed (8 keys)	2

Lattice $\varphi\mathbb{Z}^3$, cell volume $\varphi^3 = 2 + \sqrt{5}$; Icosaxes and Lotus Jewel Axes at lattice points, Unborn Axes and Dodecaxes at the cube centres $(\varphi/2)(1, 1, 1) + \varphi\mathbb{Z}^3$. Golden split point $p = m_1 + (1/\varphi)(m_2 - m_1)$.